

Reasoning for Code Changes

The changes you've provided reflect a **migration from Spring Boot 2.x to Spring Boot 3.x**, which involves several **breaking changes and API updates** due to:

- Migration from **Jakarta EE 9+** (which uses `jakarta.*` packages instead of `javax.*`)
- Major version upgrades in **Spring Security (5.7+ / 6.x)**, **Hibernate (6.x)**, and **Servlet API**
- Removal or replacement of deprecated classes and annotations

Below is a **structured explanation of why each change was made**, grouped by category:

■ 1. Package Migration: `javax.*` → `jakarta.*`

Why?

Spring Boot 3.x is based on **Jakarta EE 9+**, where all `javax.*` packages were renamed to `jakarta.*`.

Affected Files & Imports:

```
| File | Change | |-----|-----| | User.java, Book.java, BorrowRecord.java, Tag.java,
GlobalExceptionHandler.java, JwtAuthenticationEntryPoint.java,
JwtRequestFilter.java, SecurityConfig.java | All javax.persistence.*,
javax.validation.*, javax.servlet.* → jakarta.persistence.*, jakarta.validation.*,
jakarta.servlet.* |
```

■ Example:

```
import javax.persistence.*; → import jakarta.persistence.*;
import javax.servlet.http.HttpServletRequest; → import
jakarta.servlet.http.HttpServletRequest;
```

■ 2. Removed/Deprecated Classes in Spring Security

Why?

Spring Security 5.7+ (used in SB 3.x) removed `WebSecurityConfigurerAdapter`, and moved to a **functional configuration style**.

Affected File: `SecurityConfig.java`

- ■ Removed: `extends WebSecurityConfigurerAdapter`
- ■ Removed: `@EnableGlobalMethodSecurity(...)` → replaced with `@EnableMethodSecurity`

- Added: `@Bean SecurityFilterChain securityFilterChain(HttpSecurity http)` pattern
- Added: `WebSecurityCustomizer` bean for web ignore patterns

■ **Reference:** [Spring Security 6 Migration Guide](#)

■ 3. Hibernate 6.x: Removed `@Type` Annotation

Why?

Hibernate 6 removed the legacy `@Type(type = "...")` annotation in favor of `@Convert` or newer type mappings.

Affected File: `Book.java`

- Removed: `@Type(type = "yes_no")`
- Suggested fix: Use `@Convert(converter = YesNoConverter.class)` or `@JdbcTypeCode(SqlTypes.CHAR)`

■ **Note:** You may need to define custom converters if using legacy boolean flags like `'Y'/'N'`.

■ 4. Removed Duplicate / Unused Code

Some files had **duplicate imports**, **redundant fields**, or **duplicate methods**, likely from merges or refactoring errors.

Examples:

File	Issue	Fix
<code>User.java</code>	<code>import java.util.Set;</code> duplicated	Removed duplicate line
<code>User.java</code>	Two <code>prePersist()</code> methods — one without <code>@PrePersist</code> , one with	Kept only the annotated version
<code>RegisterRequestDto.java</code> , <code>BookDto.java</code> , <code>Tag.java</code>	Removed full class definitions (only kept imports or partial fields)	Likely accidental truncation — but if intentional, full DTOs should be restored

■■ **Caution:** These "partial" files (e.g., only imports in `RegisterRequestDto`) suggest possible **missing or incomplete refactoring** — verify full classes are restored.

■ 5. Validation & Bean Validation Updates

Why?

`javax.validation.constraints.*` → `jakarta.validation.constraints.*`

Affected Files:

- `RegisterRequestDto.java`, `BookDto.java`, `User.java`
- All field-level validation (e.g., `@NotBlank`, `@Email`, `@Size`) now uses `jakarta.validation` instead of `javax.validation`.

■ Note: `javax.validation` was part of Java EE; `jakarta.validation` is its EE successor.

■ 6. JPA Entity Listeners & Callbacks

Why?

Hibernate 6 changed how entity callbacks work (e.g., `@PrePersist`, `@PostLoad`), and some older patterns may break.

Affected File: `User.java`

- `prePersist()` method was defined **twice**, once without `@PrePersist`
- Only one annotated with `@PrePersist` should remain

■ 7. Additional Import Fixes

- `java.time.LocalDateTime` was duplicated in `BorrowRecord.java`, `BookDto.java`
- `java.util.Set` was duplicated in `User.java`
- Extra closing braces `}` added/removed inconsistently → likely merge artifacts

■ Summary of Key Spring Boot 3 Migration Tasks

Area	Spring Boot 2.x	Spring Boot 3.x	Package Names
Servlet API	<code>javax.*</code>	<code>jakarta.*</code>	Servlet 4 (<code>javax.servlet.*</code>) Servlet 5 (<code>jakarta.servlet.*</code>)
JPA / Hibernate	Hibernate 5.x	Hibernate 6.x	(breaking changes in <code>@Type</code> , <code>@Formula</code> , etc.)
Spring Security	<code>WebSecurityConfigurerAdapter</code> , <code>@EnableGlobalMethodSecurity</code>	<code>SecurityFilterChain</code> bean, <code>@EnableMethodSecurity</code>	
Bean Validation	<code>javax.validation</code>	<code>jakarta.validation</code>	
Lombok Compatibility	Older Lombok versions OK	Requires Lombok ≥ 1.18.22 (for SB 3.x)	

■ Recommendations for Completion

1. **Verify full DTOs & Entities:** Ensure all classes (e.g., `RegisterRequestDto`, `BookDto`) have their full field definitions restored — currently only imports remain in some diffs.
2. **Update Lombok version:** Use $\geq 1.18.22$ to support `jakarta.*` annotations.
3. **Custom Hibernate Types:** Replace `@Type(type = "yes_no")` with `@Convert` + custom `AttributeConverter`.
4. **Reintroduce Removed Security Config:** Use modern `SecurityFilterChain` bean instead of old-style `configure()`.

Let me know if you'd like help writing the full `SecurityFilterChain` bean or a `YesNoConverter` for `Book.active`.